

D799

Mobile and Ubiquitous Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A product team at a regional grocery chain with 140 stores is building a curbside-pickup app. Interviews show two clear patterns: working parents who order in short bursts on a phone, and older shoppers who order once a week on a tablet with large text. The team lead prefers a dark color theme. What is the best use of this research?

- A. Design for the average of the two groups so one layout fits everyone.
- B. Write personas from age and income data alone.
- C. Build two personas from the interview patterns and check each design decision against both.
- D. Adopt the dark theme first, since the team lead knows the product best.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A startup has a clickable prototype of its parking app. It raised \$2 million last spring, and its designers built the prototype in a popular design tool. Before any code is written, the team wants to find the biggest problems in the booking flow quickly and cheaply. What should it do?

- A. Survey 500 people on whether they like the look of the screens.
- B. Have the designers test it, since they know the design tool best.
- C. Have about five representative users try realistic booking tasks while the team watches where they struggle.
- D. Launch the app and read store reviews to find the problems.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A city transit agency with a new mobile ticketing budget has hired a design team. In the first week, a senior engineer proposes a QR-code scanner because a neighboring city uses one. Nobody on the team has spoken with riders yet. Using design thinking, what should the team do next?

- A. Complete each stage once in strict order and never return to an earlier stage.
- B. Prototype the QR scanner now, since prototyping is part of design thinking.
- C. Copy the neighboring city's app to save time.
- D. Interview and observe riders, then define the problem before choosing features.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A physical-therapy app made by a 30-person health-tech firm must measure how far a patient turns a wrist during an exercise. The phone is strapped to the forearm, and sessions take place in a clinic with good Wi-Fi. Which sensor most directly measures the turning motion?

- A. Accelerometer
- B. Gyroscope
- C. GPS receiver
- D. Wi-Fi signal strength

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A wearable must count a user's steps all day while using as little power as possible. Which sensor is the standard primary input?

- A. Accelerometer
- B. GPS, because steps are the same as distance moved.
- C. Gyroscope, because each step is a rotation of the leg.
- D. Steps can only be counted with GPS turned on at all times.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hospital wants small coin-cell tags on 400 wheelchairs so staff phones can find the nearest one within about 30 meters. Each tag must run for a year without a battery change. The hospital recently upgraded its fiber backbone. Which wireless technology best fits the tags?

- A. Bluetooth Low Energy beacons
- B. NFC
- C. Wi-Fi
- D. Wired Ethernet on the new fiber backbone

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A payment wearable must work only when the user deliberately holds it within a few centimeters of the terminal. Which technology does that range limit point to?

- A.** Cellular LTE
- B.** Wi-Fi, because the highest-bandwidth link is always the best choice.
- C.** Bluetooth Low Energy
- D.** NFC

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fitness band has a 200 mAh battery. Its heart-rate sensor and radio draw 20 mA while active and are active 10% of the time; the rest of the time the band draws 1 mA. The band's case is aluminum, and the brand sells in 14 countries. About how many hours will a full charge last?

- A.** 100 hours
- B.** About 69 hours
- C.** 200 hours
- D.** 10 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A sensor tag with a 300 mAh battery must last 100 hours. It draws 21 mA when active and 1 mA when asleep. What is the largest share of time it can be active?

- A.** 10%
- B.** 90%
- C.** 0%, because any active time drains the battery too fast.
- D.** About 14%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A running app with 2 million downloads is adding a feature that suggests nearby routes. The product manager wants to ask for location, contacts, microphone, and camera access at first launch “to save time later.” The app’s logo was just redesigned. Which approach best reflects privacy by design?

- A. Never ask for location under any circumstances.
- B. Ask only for location, at the moment the user opens route suggestions, with a clear reason.
- C. Ask for all four at launch and list them in the privacy policy.
- D. Collect everything now, since users can delete their accounts later.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A review finds that a weather app keeps every user’s precise GPS history for five years, although it only shows city-level forecasts. Which principle does this most directly violate?

- A. Availability
- B. Encryption strength
- C. Data minimization
- D. None, because users always accept this by agreeing to the terms.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A note-taking app syncs across phone, tablet, and laptop. A user edits the same note on the phone during a flight with no signal and on the laptop at home; the laptop has a 16-inch screen and the phone uses a dark theme. When the phone reconnects, which design best protects the user’s work?

- A. Keep whichever version synced last and discard the other silently.
- B. Block all editing whenever a device is offline.
- C. Sync only between devices with the same screen size and theme.
- D. Save edits locally, queue them, and on reconnect detect the conflict and merge or show both versions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A product team at a regional grocery chain ~~with 140 stores~~ is building a curbside-pickup app. Interviews show two clear patterns: working parents who order in short bursts on a phone, and older shoppers who order once a week on a tablet with large text. ~~The team lead prefers a dark color theme.~~ What is the best use of this research?

Target word: how to turn research into a design tool **Decides:** Two distinct goal-and-context patterns emerged from interviews.

A	Adjacent category	An average is a statistic, not a persona; it often fits neither group.
B	Dropped element	Drops goals, behaviors, and context — the parts that make personas useful.
C	Correct	Personas are research-based archetypes; two distinct patterns call for two personas.
D	Camouflage	Answers the lead's taste, which does not decide user needs.

2. Answer C Concept 2

A startup has a clickable prototype of its parking app. ~~It raised \$2 million last spring,~~ and its designers built the prototype ~~in a popular design tool.~~ Before any code is written, the team wants to find the biggest problems in the booking flow quickly and cheaply. What should it do?

Target word: fast, cheap way to find usability problems **Decides:** Pre-code prototype and the goal of finding problems, not measuring preference.

A	Confusion set	Measures stated preference, not whether people can complete tasks.
B	Camouflage	Designers are not representative users; tool skill does not decide the method.
C	Correct	Small formative tests of real tasks reveal most major problems early and cheaply.
D	Inverted operation	Tests after release, when fixes cost the most.

3. Answer D Concept 3

A city transit agency ~~with a new mobile ticketing budget~~ has hired a design team. In the first week, a senior engineer proposes a QR-code scanner ~~because a neighboring city uses one.~~ Nobody on the team has spoken with riders yet. Using design thinking, what should the team do next?

Target word: next step in the design thinking cycle **Decides:** No user research has been done yet.

A	Absolute language	Design thinking is iterative; teams loop back as they learn.
B	Dropped element	Skips empathize and define; prototypes should test ideas grounded in user needs.
C	Camouflage	Another city's choice does not reveal this city's riders' problems.
D	Correct	Empathize and define come before ideating and prototyping.

4. Answer B Concept 4

A physical-therapy app ~~made by a 30-person health-tech firm~~ must measure how far a patient turns a wrist during an exercise. The phone is strapped to the forearm, and sessions take place in a clinic ~~with good Wi-Fi~~. Which sensor most directly measures the turning motion?

Target word: sensor for rotation **Decides:** The quantity is rotation of the forearm.

A	Confusion set	Measures linear acceleration; it can sense tilt but not a clean rotation about the forearm.
B	Correct	A gyroscope measures rotation rate, which integrates to the angle turned.
C	Adjacent category	Gives location, not orientation, and works poorly indoors.
D	Camouflage	Answers the Wi-Fi detail; it estimates position, not wrist rotation.

5. Answer A Concept 4 • transfer

A wearable must count a user's steps all day while using as little power as possible. Which sensor is the standard primary input?

Target word: reverse: from the task to the sensor **Decides:** Steps show up as repeated linear acceleration peaks.

A	Correct	Step counting detects the rhythmic acceleration peaks of walking at very low power.
B	Adjacent category	GPS measures location, fails indoors, and is one of the biggest power drains.
C	Confusion set	Measures rotation rate and draws more power; it is a helper, not the primary step input.
D	Absolute language	Accelerometer-based step counting works with GPS off.

6. Answer A Concept 5

A hospital wants small coin-cell tags on 400 wheelchairs so staff phones can find the nearest one within about 30 meters. Each tag must run for a year without a battery change. ~~The hospital recently upgraded its fiber backbone.~~ Which wireless technology best fits the tags?

Target word: radio for low-power, room-scale tags **Decides:** Coin cell for a year plus about 30 meters of range.

A	Correct	BLE is built for tiny power budgets and ranges of tens of meters, and phones read it natively.
B	Confusion set	Works only within a few centimeters, so staff could not find chairs across a room.
C	Adjacent category	Right range, but its power draw would drain a coin cell in days or weeks.
D	Camouflage	Answers the backbone detail; wheelchairs cannot be wired.

7. Answer D Concept 5 • transfer

A payment wearable must work only when the user deliberately holds it within a few centimeters of the terminal. Which technology does that range limit point to?

Target word: reverse: from the range requirement to the radio **Decides:** A few centimeters.

A	Adjacent category	A wide-area link, not a proximity link.
B	Absolute language	Bandwidth is not the requirement; range and intent are.
C	Confusion set	Reaches tens of meters, so a payment could trigger without a deliberate tap.
D	Correct	NFC's centimeter range makes the tap a deliberate act.

8. Answer B Concept 6

A fitness band has a 200 mAh battery. Its heart-rate sensor and radio draw 20 mA while active and are active 10% of the time; the rest of the time the band draws 1 mA. ~~The band's case is aluminum, and the brand sells in 14 countries.~~ About how many hours will a full charge last?

Target word: battery life in hours **Decides:** Average current = $0.10 \times 20 + 0.90 \times 1 = 2.9$ mA.

A	Dropped element	$200 \div 2$ counts only the active draw and ignores the 1 mA sleep current.
B	Correct	$200 \text{ mAh} \div 2.9 \text{ mA} \approx 69$ hours.
C	Wrong stem number	$200 \div 1$ uses only the sleep current.
D	Dropped element	$200 \div 20$ assumes the band is active 100% of the time.

9. Answer A Concept 6 · transfer

A sensor tag with a 300 mAh battery must last 100 hours. It draws 21 mA when active and 1 mA when asleep. What is the largest share of time it can be active?

Target word: solve for duty cycle **Decides:** Allowed average current = $300 \div 100 = 3$ mA.

A	Correct	$1 + 20d = 3$, so $d = 0.10$.
B	Inverted operation	That is the share of time asleep, not active.
C	Absolute language	The budget allows a 10% duty cycle.
D	Dropped element	$3 \div 21$ ignores the sleep current during the other 86%.

10. Answer B Concept 7

A running app ~~with 2 million downloads~~ is adding a feature that suggests nearby routes. The product manager wants to ask for location, contacts, microphone, and camera access at first launch "to save time later." ~~The app's logo was just redesigned.~~ Which approach best reflects privacy by design?

Target word: privacy-by-design permission approach **Decides:** The feature needs only location.

A	Absolute language	The feature legitimately needs location; the principle is necessity, not prohibition.
B	Correct	Minimizes data and asks in context for what the feature actually uses.
C	Dropped element	Disclosure alone drops data minimization; contacts, mic, and camera are not needed.
D	Inverted operation	Pushes privacy to cleanup after the fact instead of building it in.

11. Answer C Concept 7 · transfer

A review finds that a weather app keeps every user's precise GPS history for five years, although it only shows city-level forecasts. Which principle does this most directly violate?

Target word: reverse: from the practice to the principle **Decides:** Data kept is far more precise and longer-lived than the purpose needs.

A	Confusion set	Availability is about access to systems, not over-collection.
B	Adjacent category	Encrypting the data would not make keeping it necessary.
C	Correct	Collect and retain only what the stated purpose needs.
D	Absolute language	Agreeing to terms does not make unneeded collection a good practice.

12. Answer D Concept 8

A note-taking app syncs across phone, tablet, and laptop. A user edits the same note on the phone during a flight with no signal and on the laptop at home; ~~the laptop has a 16-inch screen and the phone uses a dark theme.~~ When the phone reconnects, which design best protects the user's work?

Target word: sync design for offline conflicting edits **Decides:** Two devices changed the same note while one was offline.

A	Dropped element	Drops conflict detection, so one set of edits is lost.
B	Inverted operation	Makes the app online-only instead of designing for intermittent connectivity.
C	Camouflage	Display settings do not affect data conflicts.
D	Correct	Offline-first sync with conflict detection loses no work.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Personas come from research	A persona is an archetype built from observed user goals, behaviors, and context, used to test design decisions.	Designing for an average user, a demographic label, or the team's own taste.	Interview findings that fall into distinct patterns of goals or context.
2	Usability testing observes behavior	Small formative tests with representative users doing realistic tasks surface the biggest problems early and cheaply.	Asking people whether they like a design, or waiting for launch to learn.	“Before coding” or “find problems quickly” with a prototype in hand.
3	Design thinking starts with empathy	Empathize and define the problem before ideating and prototyping; the cycle is iterative, not a one-way checklist.	Jumping to a feature someone already likes.	A proposed solution appears before anyone has talked to users.
4	Matching sensors to what they measure	Accelerometer = linear acceleration (steps, shakes, tilt from gravity); gyroscope = rotation rate; magnetometer = heading; GPS = outdoor location.	Treating any motion sensor as interchangeable.	The physical quantity named in the requirement: rotate, move, face, or where.
5	Connectivity trade-offs	Choose the radio by range, power, and bandwidth: NFC = centimeters, BLE = low power over meters, Wi-Fi = high bandwidth at higher power, cellular = wide area.	Always picking the fastest link.	A battery limit or a distance limit in the requirement.
6	Duty cycle and battery life	Battery life (hours) = capacity (mAh) ÷ average current (mA), where average current weights active and sleep draw by the time spent in each.	Dividing by the peak current or ignoring the sleep current.	An active draw, a sleep draw, and a percentage of time active.
7	Privacy by design	Collect and keep only the data a feature needs, and ask for each permission in context when the feature is used.	Requesting everything up front and relying on the privacy policy.	A permission list longer than the feature being built.
8	Offline-first multi-device sync	Store changes locally, queue them, and detect and resolve conflicts on reconnect so no user work is lost.	Assuming devices are always online or letting the last write silently win.	The same record edited on two devices while one was disconnected.

Course scope (public description): Mobile and Ubiquitous Design blends in-depth technical knowledge of computing systems like sensors, processors, embedded systems design, communication protocols, and connectivity technologies with user-centered design principles for creating functional and engaging mobile applications. The course will demonstrate how to apply design thinking and UX methods such as user personas, usability testing, and addressing real-world challenges to develop scalable, secure, and ethical solutions that optimize the interplay between various devices. By emphasizing privacy considerations and the ability to adapt across different user environments, this course prepares students to become proficient in analyzing user needs and designing synchronized multi-device systems through mobile apps that effectively meet those requirements.